

Problem 1

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int arr[8] = {10, 20, 30, 40, 50, 60, 70, 80};
```

```
    cout << "Elements in ARRAY: ";
```

```
    for (int i = 0; i < 8; i++)
```

```
    {
```

```
        cout << arr[i] << " ";
```

```
    }
```

```
    cout << endl;
```

```
    return 0;
```

```
}
```

```
task1 > problem1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int arr[8] = {10, 20, 30, 40, 50, 60, 70, 80};
7
8      cout << "Elements in ARRAY: ";
9      for (int i = 0; i < 8; i++)
10     {
11         cout << arr[i] << " ";
12     }
13     cout << endl;
14
15     return 0;
16 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks> cd "c:\Users\Uzair\Downloads\"
) { .\problem1 }
Elements in ARRAY: 10 20 30 40 50 60 70 80
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task1>
```

Problem 2

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

```
{
```

```
    int size = 8;
```

```
    int arr[size] = {10, 20, 30, 40, 50, 60, 70, 80};
```

```
int pos = 0;

for (int i = size; i > pos; i--)
{
    arr[i] = arr[i - 1];
}

arr[pos] = 5;

size++;

cout << "After Insertion in ARRAY: ";

for (int i = 0; i < size; i++)
{
    cout << arr[i] << " ";
}

cout << endl;

return 0;

}
```

```
Welcome X problem1.cpp X problem2.cpp X
task1 > G+ problem2.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int size = 8;
7      int arr[size] = {10, 20, 30, 40, 50, 60, 70, 80};
8
9      int pos = 0;
10
11     for (int i = size; i > pos; i--)
12     {
13         arr[i] = arr[i - 1];
14     }
15     arr[pos] = 5;
16     size++;
17
18     cout << "After Insertion in ARRAY: ";
19     for (int i = 0; i < size; i++)
20     {
21         cout << arr[i] << " ";
22     }
23 }
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks> cd "c:\Users\Uzair\Downloads\semester\Semester 4"
) { .\problem2 }
After Insertion in ARRAY: 5 10 20 30 40 50 60 70 80
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task1>
```

Problem 3

```
#include <iostream>
```

```
using namespace std;
```

```
int main()
```

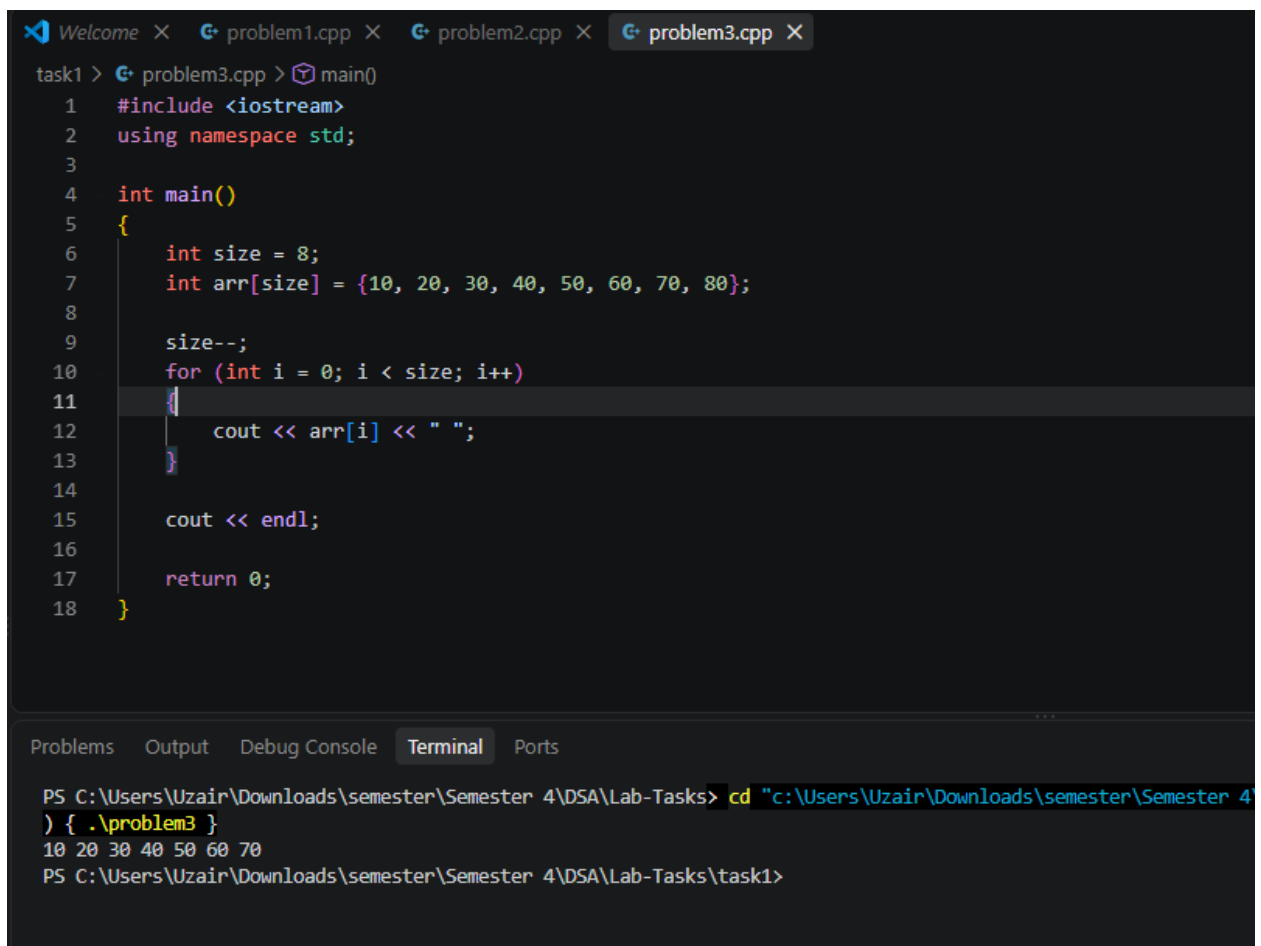
```
{
```

```
    int size = 8;
```

```
    int arr[size] = {10, 20, 30, 40, 50, 60, 70, 80};
```

```
    size--;
```

```
for (int i = 0; i < size; i++)  
  
{  
  
    cout << arr[i] << " ";  
  
}  
  
cout << endl;  
  
return 0;  
  
}
```



The screenshot shows a C++ IDE with three tabs: 'Welcome', 'problem1.cpp', and 'problem2.cpp'. The active tab is 'problem3.cpp'. The code in the editor is as follows:

```
task1 > G+ problem3.cpp > main()  
1  #include <iostream>  
2  using namespace std;  
3  
4  int main()  
5  {  
6      int size = 8;  
7      int arr[size] = {10, 20, 30, 40, 50, 60, 70, 80};  
8  
9      size--;  
10     for (int i = 0; i < size; i++)  
11     {  
12         cout << arr[i] << " ";  
13     }  
14  
15     cout << endl;  
16  
17     return 0;  
18 }
```

The IDE has a bottom panel with tabs for 'Problems', 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active, showing the following command prompt session:

```
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks> cd "c:\Users\Uzair\Downloads\semester\Semester 4"  
> { .\problem3 }  
10 20 30 40 50 60 70  
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task1>
```

Problem 4

```
#include <iostream>

using namespace std;

int main()

{

    int size = 8;

    int arr[size] = {10, 20, 30, 40, 50, 60, 70, 80};


    int pos = 0;


    cout << "Elements in ARRAY: ";

    for (int i = 0; i < size; i++)

    {

        cout << arr[i] << " ";

    }


    cout << endl;


    for (int i = size; i > pos; i--)

    {

        arr[i] = arr[i - 1];

    }

    arr[pos] = 5;

    size++;
```

```
cout << "After Insertion in ARRAY: ";

for (int i = 0; i < size; i++)

{

    cout << arr[i] << " ";

}


cout << endl;


for (int i = pos; i < size; i++)

{

    arr[i] = arr[i + 1];

}

size--;


cout << "After Deletion in ARRAY: ";

for (int i = 0; i < size; i++)

{

    cout << arr[i] << " ";

}


cout << endl;


return 0;

}
```

task1 > Problem4.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      int size = 8;
7      int arr[size] = {10, 20, 30, 40, 50, 60, 70, 80};
8
9      int pos = 0;
10
11     cout << "Elements in ARRAY: ";
12     for (int i = 0; i < size; i++)
13     {
14         cout << arr[i] << " ";
15     }
16
17     cout << endl;
18
19     for (int i = size; i > pos; i--)
20     {
21         arr[i] = arr[i - 1];
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks> cd "c:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task1"
) { .\Problem4 }
Elements in ARRAY: 10 20 30 40 50 60 70 80
After Insertion in ARRAY: 5 10 20 30 40 50 60 70 80
After Deletion in ARRAY: 10 20 30 40 50 60 70 80
PS C:\Users\Uzair\Downloads\semester\Semester 4\DSA\Lab-Tasks\task1>
```